MV-22B Osprey

Flight Manual

for the FlightGear MV-22B with its JSBSim flight model

This manual describes the aircraft as simulated: its flight model, cockpit, displays and automatic flight control system. Much of it has no counterpart in the real aircraft, whose manual is not public, and none of it is the real aircraft's manual or procedures. It is generated with the aircraft: its numbers, tables and charts come from the aircraft's own data, so they follow it as it changes.

Generated from the aircraft's sources (`manual/` in its repository).

How to use this manual

The manual is in parts, as a military flight manual is: the aircraft and its systems (Part I), its limitations (Part II), normal procedures (Part III), flight characteristics (Part IV), emergency procedures (Part V) and performance data (Part VI). Keys are given as `m` or `Ctrl-T`; words the displays show, as HVR CPLD HOLD.

Warnings, cautions and notes. Three kinds of notice stand apart from the text, by how much they matter:

WARNING

An operating procedure or technique that, not followed, may lose the aircraft.

CAUTION

An operating procedure or technique that, not followed, may damage the aircraft or end the flight's purpose.

Note

An operating procedure, technique or fact worth emphasising.

Wording. *Shall* is mandatory; *should* is recommended; *may* and *need not* are optional; *will* states what happens, not a requirement.

Its numbers. Weights are pounds, speeds knots (KCAS, calibrated, unless stated), heights feet, power shaft horsepower. Performance is the flight model's: the comparisons that support it are in the model's report.

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Part I

The aircraft

Chapter 1

General description

1.1 The aircraft

The MV-22B is a tiltrotor: two proprotors, 38 ft across, at the tips of a wing, on nacelles that tilt from vertical (90°, helicopter mode) to horizontal (0°, airplane mode) and 97.5−90 = 7.5° aft of vertical. In helicopter mode the rotors lift the aircraft and it hovers and manoeuvres as a helicopter does; in airplane mode the wing lifts it and the rotors are propellers. Between them it *converts*, the nacelles moving at up to 8°/s within a corridor of nacelle angle against airspeed (Part II).

Two engines drive the rotors through gearboxes and an interconnect shaft: either engine turns both rotors. The flight controls mix the pilot's controls into each mode's: the rotors' collective and cyclic in helicopter mode, the elevator, flaperons and rudders in airplane mode, both through the conversion. A fly-by-wire augmentation holds attitude in every mode, and an automatic flight control system (AFCS) flies the aircraft when coupled, down to a hover and a landing.

1.2 Weights

Empty	33 140
At the start: three crew, tanks half full, no cargo	38 730
Reference loading (maximum vertical take-off weight of the 1986 design)	47 500
Fuel, full (1 448 US gal)	9 860
Troops and cargo in the cabin, up to	20 000

Fuel and cargo are set from Equipment → Fuel and Payload; the Troops and cargo slider places the load in the cabin. 24 equipped troops weigh about 5800 lb.

Chapter 2

Systems

2.1 Engines and drive

Each engine is rated at 6 150 shp (maximum rated power; intermediate 5 904, maximum continuous 4 982), falling with altitude and temperature. The thrust control lever sets both engines' power demand from flight idle to the rating. The drive limits the torque: 9 100 hp at 100 % rotor speed, 7 644 hp at 84 %, shared between the running engines, so that near sea level the drive, not the engines, limits the power. A governor holds the rotor speed (N_R) with the rotors' collective: 104 % in helicopter mode and through the conversion, 84 % in airplane mode with the nacelles at 0°. A topping governor takes power off above 2 % of overspeed.

The ENG format shows the alerts (Section 3.4), each engine's torque, gas generator speed and fuel flow, the rotor speed and the drive's torque limit.

2.2 Fuel

Five tanks, drawn in the transfer sequence's order: the right aft sponson first, then the forward sponsons, the feed tanks last.

Tank	US gal	lb
Left feed	88	599
Right feed	88	599
Left forward sponson	478	3255
Right forward sponson	478	3255
Right aft sponson	316	2152
total	1448	9861

The FUEL format shows the tanks, the flow and the endurance; the NAV format the endurance and range. FUEL LOW shows below 1000 lb.

2.3 Flight controls

Thrust control lever (TCL). The throttle axis, or `Page Up`/`Page Down`. Up is more collective in helicopter mode and more power in airplane mode. At the start's weight a hover needs about two thirds of it, at the reference loading nearly all.

Nacelles. `m` tilts the nacelle command 10° forward, `M` 10° aft; the NACELLE thumbwheel on either lever does the same (click it, or the mouse wheel over it). The nacelles follow the command at the conversion rate. From 90° `M` tilts them to 97.5°, 7.5° aft of vertical, to stop or back up in the hover.

Conversion protection. The command is held within the corridor (Part II): below 35 kt the nacelles go no further forward than 75°, and as the aircraft accelerates they follow the corridor's lower edge, so a command fully forward converts as the speed allows. Slowing below the corridor for the present angle, the protection raises them. CONV PROT shows while it holds the command off the pilot's.

Stick and pedals. The elevator and aileron axes and the rudder axis. The flight controls mix them into the rotors' cyclic and differential collective in helicopter mode, phased with the nacelles, and into the elevator, flaperons and rudders in every mode.

Flaperons. Scheduled by the flight model: 67° in the hover (least download), 40° through the conversion and in airplane mode below 120 kt, retracted by 200 kt. The flap lever on the console does nothing.

2.4 Flight control augmentation

The augmentation holds the aircraft's attitude in all modes. Its response type changes with speed:

- *Below 50 kt with the nacelles above 60°*: attitude command, heading hold. Each 5 % of stick (one arrow-key press) holds about 1° of pitch or 1.25° of bank; each 5 % of pedal turns the aircraft at 1.5°/s (30°/s at full travel), and the heading holds with the pedals centred.
- *Above it*: rate command, attitude hold. The stick commands a rate, and the aircraft holds the attitude it is released at.

It switches at 55 kt accelerating and 45 kt slowing, so that a speed held near 50 kt does not switch back and forth; both share one reference, so the switch is free of transients. A trim schedule keeps the stick's centre trimmed through the conversion: hands off, the aircraft holds its attitude through it.

2.5 Landing gear, brakes and the wing

g/G raise and lower the gear; GEAR DOWN shows with the gear down above 250 KCAS and 50 ft. B sets the parking brake. The rotor brake holds the stopped rotors (ROTOR BRAKE); the start releases it. N folds the blades and swings the wing along the fuselage, n unfolds it; the rotors shall be stopped.

2.6 Doors, lights and others

d crew door, D cargo doors, e steps, l landing light. The complete list of keys is in Table A.1.

Chapter 3

Cockpit and displays

3.1 Layout

Four multifunction displays (MFD 1 to 4 from the left) across the instrument panel; the control display unit (CDU) and its two keypads on the centre console with the standby engine and fuel display (SFD) above it; the electronic standby flight instrument (ESFI); the flight control panel (the AFCS's keys and knobs) and the radio panels on the glareshield; a thrust control lever beside each seat.

3.2 Multifunction displays

The keys along each display's top pick its format: FLT, NAV, ENG, FUEL. A format's own keys, down its right side, show their labels for a few seconds after a key is pressed.

Flight (FLT). One format for every regime (Figure 3.1). At its centre the flight path marker, where the aircraft is going, with its acceleration chevron; airspeed and altitude tapes with 10 s trend vectors; the heading; a ground grid in perspective. The limits are drawn where the eye already is: the conversion corridor's airspeeds for the present nacelle angle on the airspeed tape, SINK RATE's boundary on the vertical speed scale (its bar turning amber, then red, as the descent nears it), and the margin to the first power limit as one bar, with the power to hover in and out of ground effect marked. A small corridor diagram shows where the aircraft is in the conversion and where it is going. The AFCS's mode line runs along its top (Section 4.3).



1. AFCS mode line: the modes, green when coupled
2. airspeed tape, KIAS, with its 10 s trend; the red bands: the conversion corridor's limits for the nacelle angle
3. airspeed
4. flight path marker
5. acceleration chevron: above the marker accelerating, below slowing
6. altitude, with its tape and trend
7. vertical speed scale; the bar coloured by SINK RATE's boundary
8. heading
9. ground speed
10. radar altitude
11. nacelle angle in the conversion corridor: where the aircraft is (circle) and its limits
12. power: torque against the first limit; OGE and IGE: the power to hover out of and in ground effect
13. rotor speed
14. gear, flaperons and the augmentation's response type

Figure 3.1: The flight format in airplane mode, coupled at 230 kt.

Slowing below 30 kt with the nacelles past 75° it turns to its *hover format* (Figure 3.2): the flight path marker gives way to the drift seen from above, a ball on a line from the centre at the ground velocity's opposite 3 s ahead, to fly the stick towards; the hover hold's point, a magenta box filled once it holds it; the radar height. Below 200 ft the height tape gives way to a green bar under the readout that drains to its foot as the height goes, at the rate the aircraft closes on the ground. Past 60° of bank or 25° of pitch the clutter goes and an arrow shows the way to roll level. R7 (HVR MARK) marks the present position as a hover point.

With both engines out (ENG OUT, Section 4.4.12) the format flies that mode, coupled or not: the guidance cue is its pitch and bank, the speed bug (magenta) its glide speed, the corridor's target its nacelles, TCL UP and TCL DN its lever. The altitude tape marks the flare's height (FLR). In the flare a red line on the pitch ladder (TAIL) marks the pitch at which the fuselage meets the ground on the wheels. In place of the configuration line, in amber, the glide's reach along the track to the flare (REACH 6.2 NM; CONVERTING until airplane mode).



1 AFCS mode line: the hover hold's state (DECEL, HOLD, TRC) and height hold

2 the hover point: the hold's stop, filled once it holds it; its distance

3 the drift ball, where the aircraft is drifting 3 s on, to fly the stick towards

4 the ring at 10 kt of drift

5 radar height

6 below 200 ft: the ground closing, a bar that drains as the height goes

7 barometric altitude

8 the high sink rate limit for the airspeed

9 ground speed

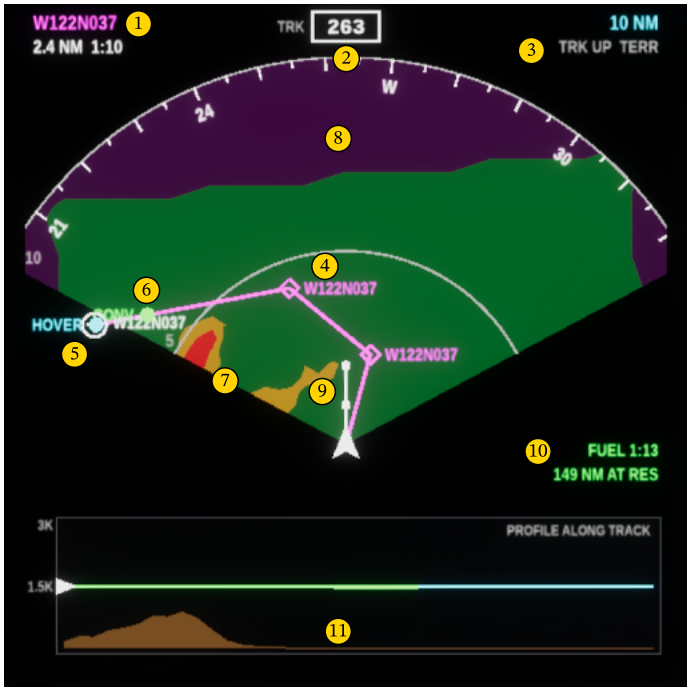
10 nacelle angle in the conversion corridor

11 power, with the power to hover out of and in ground effect

12 the augmentation's response type: attitude command

Figure 3.2: The flight format's hover format, the hover hold holding 150 ft over the sea.

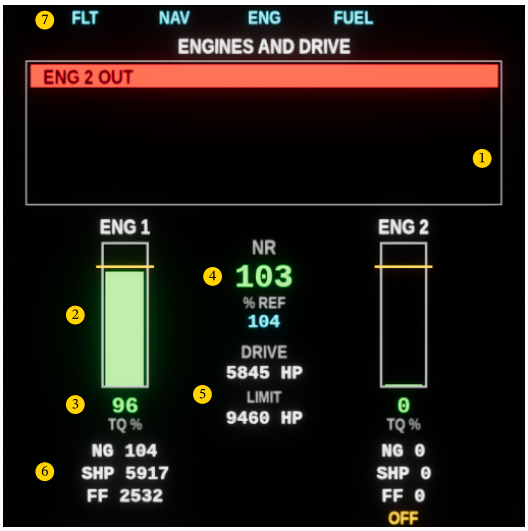
Navigation (NAV). A moving map, track up (R3: north up; Figure 3.3). The terrain is coloured by its clearance below the height the aircraft will have when it gets there (60 s ahead at most; below 90 kt in the helicopter's tighter bands); where the scenery has no elevation yet it is dim magenta, and below 100 ft above the ground it is measured from 100 ft up, so that flat ground around a hover stays quiet. On it: the route and its waypoints; the landing zone, the route's last point, with its approach into the wind; the conversion planner's points, where to begin slowing down (DECEL) and converting (CONV) to arrive in a hover; the path 30 and 60 s ahead in the turn; an approach line; traffic; the fuel's endurance and range. Below it the profile along the track: the terrain, the aircraft's path, the selected altitude. R1/R2 change the range, R4 the terrain. With ENG OUT on, gliding, an amber dashed circle (GLIDE) is where the glide reaches the flare's height: from the energy height (the height and the speed over the glide's) at the model's own glide ratio for the weight, the gear coming down from 1000 ft if it is still up, centred where the wind carries the air over the glide's time; the profile shows the glide's line to it. It errs short, by 4–5 % at the reference loading and 8–12 % at FlightGear's start loading: the engines' run-down after a cut and the speed's sag as the gear comes are left out.



1 next waypoint: distance and time to it
2 track
3 range; track or north up; terrain shown
4 the route
5 the landing zone, the route's last point, with its approach into the wind
6 the conversion planner: where to begin converting (CONV) and slowing (DECEL)
7 terrain coloured by its clearance below the aircraft's height when it gets there
8 no elevation yet from the scenery
9 the path 30 and 60 s ahead
10 fuel: endurance and range at the reserve
11 the profile along the track: terrain, the path, the selected altitude

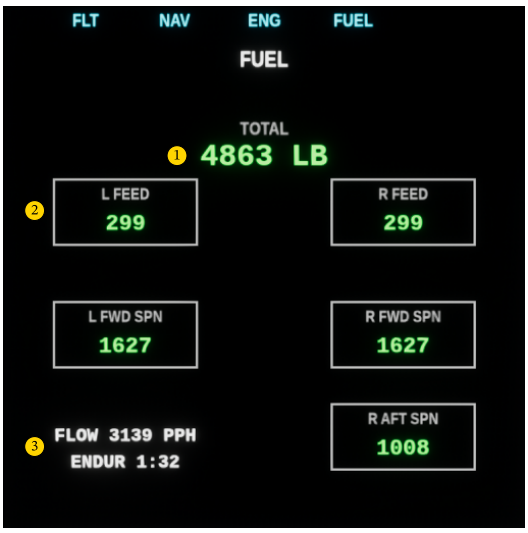
Figure 3.3: The navigation format with a route over the peninsula's ridge to a landing zone on the coast, at 120 kt: the terrain beyond the coast not yet loaded by the scenery.

Engines (ENG) and **fuel (FUEL)**: the engines and drive, the tanks and the flow (Figures 3.4 and 3.5).



1 the alerts, the latest first in each level; one not yet acknowledged in reverse video
2 each engine's torque against its limit (the line)
3 torque, %
4 rotor speed and its reference
5 the drive's power and its limit
6 gas generator speed, shaft power, fuel flow
7 the formats' keys

Figure 3.4: The engines and drive format, engine 2 shut down in flight: ENG 2 OUT not yet acknowledged.



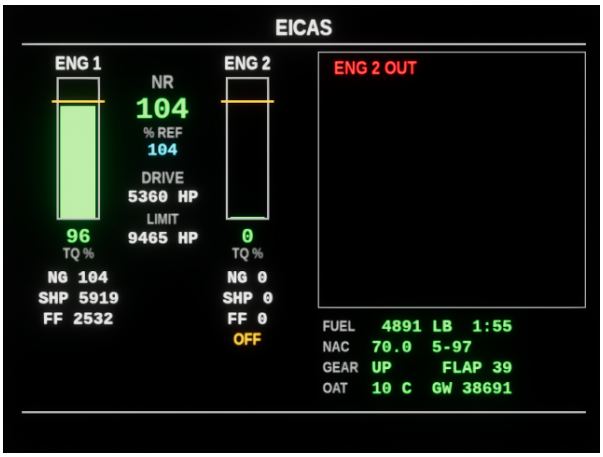
1 total fuel
2 the five tanks
3 fuel flow and endurance

Figure 3.5: The fuel format.

3.3 Control display unit

The CDU shows the EICAS page unless another is called up (Figure 3.6a): on the left the engines and the drive as the ENG format has them, on the right the alerts (Section 3.4), under them the fuel with its endurance, the nacelle angle with the conversion corridor's range at the present speed, the gear and the flap, the outside air temperature and the gross weight, each amber while its alert stands. Its keys B4, B2, B3 and B1 show the EICAS page, the C/A page (the alerts cleared, the latest first and how long ago), the STAT page (weights) and the scratchpad's page; B5 (ACK) acknowledges the alerts. The keypads type into the scratchpad; CLR clears it, DEL deletes a character, ENTER enters it:

Entry	Sets
`H270`	the heading bug
`C090`	NAV 1's course
`B2992`	the altimeter (inches of mercury × 100)
`T300`	TF's clearance, 100–1000 ft
`R1`, `R2`, `R3`	TF's ride: soft, medium, hard



(a) the EICAS page, engine 2 shut down in flight, `ENG 2 OUT` acknowledged



(b) the status page

Figure 3.6: The control display unit.

3.4 Alerts

Three levels, as MIL-STD-411F has them for military cockpits: warnings red, cautions amber, advisories green. The alerts standing show in the window at the top of the ENG format (C/A under any multifunction display calls it up) and in the one on the CDU's EICAS page: warnings first, then cautions, then advisories, the latest first within each, one not yet acknowledged in reverse video; a window's last line counts the ones that do not fit.

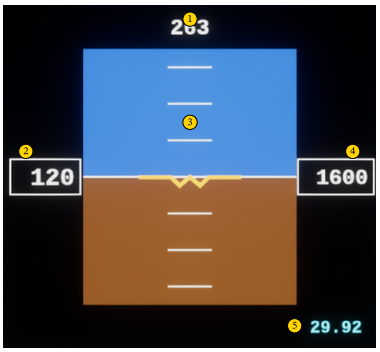
A new warning lights MASTER WARN on the master alert at each end of the glareshield (the plain square key of the pair), sounds the warning tone and says the warning in FlightGear's own voice, both again every 4 s; a new caution lights MASTER CAUT and chimes once. The tone and the voice stop after 10 s; the master stays lit until the alert clears or is acknowledged: press either master or ACK under any multifunction display or on the CDU. On the ground with the rotors carrying under half the weight (starting, taxiing, shutting down) the alerts are listed only, without the masters or a sound; one still standing at lift-off becomes new then. An autopilot mode that the aircraft lets go of (not the pilot) is listed amber with its own warbling tone until acknowledged or engaged again, and shows on the flight format for 5 s. A condition holds for 1 s before its caution or OVERSPEED shows, so that a value on its threshold does not chime twice.

Three warnings are time-critical: SINK RATE, PULL UP and ROTOR RPM. Each shows in red in the flight format's attitude window, under the aircraft, and says itself every 2.5 s for as long as it stands; acknowledging puts out the master but not the voice. SINK RATE is the vortex ring state ahead: either rotor descending past the boundary the flight format's vertical speed scale marks, which is the high sink rate limit's 800 ft/min in vertical descent at the reference loading at sea level (Section 5.4), and moves with the weight, the air's density and the edgewise speed as the state's onset does. With the lever drawn back steadily it comes 6–8 s before the onset (at 0.02 of the lever a second), and 12–15 s at half that rate. ROTOR RPM: the rotor speed so low that the rotors at their greatest thrust no longer hold the hover. PULL UP: with the nacelles down, descending to the ground in less than a reaction and a pull-out; not with ENG OUT on, which has no power to climb away and whose own flare closes on the ground quicker than that. Both engines out is one warning, DUAL ENG OUT, collecting ENG 1 OUT and ENG 2 OUT.

Alert	Level	When
SINK RATE	warning (red)	either rotor descending past 0.39 of the vortex ring state's onset for its edgewise speed (Johnson's model, in its own hover induced velocity): 800 ft/min in vertical descent at the reference loading at sea level
PULL UP	warning (red)	nacelles under 60 deg, descending to the ground in less than 5 s and a 0.5 g pull-out; not with ENG OUT on (no power to climb away: the descent to the ground is the mode's, its flare closing on the ground quicker than 5 s)
ROTOR RPM	warning (red)	nacelles past 60 deg: the rotor speed drooping, too low for the rotors' greatest thrust to hold the hover at this weight and density
DUAL ENG OUT	warning (red)	both engines that ran stopped (ENG 1 OUT and ENG 2 OUT collected into it)
ENG 1 OUT	warning (red)	an engine that ran stops, the other running (both: DUAL ENG OUT; both shut down on the ground, the alerts clear)
ENG 2 OUT	warning (red)	an engine that ran stops, the other running (both: DUAL ENG OUT; both shut down on the ground, the alerts clear)
OVERSPEED	warning (red)	past the conversion corridor's upper speed for the nacelles' angle; in airplane mode 280 KCAS or Mach 0.48, 220 KCAS above 94 % rotor speed
FLAPPING	warning (red)	a gimbal at its flapping stop (12 deg)
WHEELS	warning (red)	nacelles past 60 deg under 200 ft, descending, the gear not down
FUEL LOW	caution (amber)	under 1000 lb of fuel
NR	caution (amber)	the rotor speed more than 3 % off its reference; not in ENG OUT's settling and after (its lever spends the rotor speed in the last 20 ft)
HOVER PWR	caution (amber)	slower than 60 KEAS, nacelles past 60 deg: the power to hover at this height more than the engines running have
FLAPPING HIGH	caution (amber)	a gimbal past 10.5 deg of flapping, the wing clearance's design value
AIRSPEED LOW	caution (amber)	slower than the conversion corridor's least speed for the nacelles' angle
CONV PROT	caution (amber)	the conversion protection holding the nacelle command off the pilot's
GEAR DOWN	caution (amber)	the gear down above 250 KCAS (an assumed limit) and 50 ft
ROTOR BRAKE	advisory (green)	the rotor brake on
BATTERY OFF	advisory (green)	the battery switch off
HVR CPLD OFF	autopilot (amber)	the hover coupler let go
HVR CPLD UNAVAIL	advisory (green)	the hover coupler refused (listed for 10 s)
CPLD OFF	autopilot (amber)	the coupling let go, by the stick or on the ground
AUTO NAC OFF	autopilot (amber)	AUTO NAC let go

3.5 Standby instruments and lighting

The ESFI shows attitude, airspeed and altitude, the SFD the engines and fuel (Figure 3.7). The displays need power, the battery or an engine running. They follow the light outside as their photocells do: DAY is bright enough for full sun and dims at dusk, NIGHT stays at a night vision goggle level. The standby instrument, the flight control panel's windows and the radio panels have no NIGHT/DAY knob and follow the light over both. The BRT rockers and knobs scale each.



(a)

1 heading
2 airspeed
3 attitude
4 altitude
5 altimeter setting



(b)

Figure 3.7: The standby instruments: (a) the electronic standby flight instrument, (b) the standby engine and fuel display.

Chapter 4

Automatic flight control system

4.1 Description

The AFCS is a flight director that the augmentation flies when *coupled*. Its modes compute attitudes, the lever and the nacelles; with CPLD on, the augmentation flies them, and with it off they remain cues. Moving the stick out of its centre uncouples it (CPLD OFF); CPLD couples it again. The coupled AFCS drives the lever while a speed mode is on: moving the lever flies through it, and left alone for 1.5 s the AFCS takes it again.

Speed and the flight path share the lever and the pitch as total energy control does: in airplane mode the lever flies the total energy's rate (climb and acceleration together) and the pitch shares it between climb and speed. Blended by the nacelle angle from 50 to 80°, on the back side of the power curve the lever flies the climb and the pitch the speed, as a powered-lift aircraft is flown there.

4.2 Controls

The flight control panel's keys and knobs (Table 4.1), or the keys (Table A.1). A hold key takes the present heading, speed or altitude onto its knob when pressed; the knob then changes it. The knobs' values show in the panel's windows.

Table 4.1: The flight control panel.

Key or knob	Function
CPLD	Couples the flight director to the augmentation (either key)
ENAV	Not installed
VOR/ILS	VOR/ILS: NAV 1's localizer and glide slope, or its VOR
TACAN	Not installed
SRCH PAT	Not installed
APPR	Approach to a hover; again: land; again: off
WYPT	The route manager's route
HDG HOLD	Heading hold (HDG knob)
SPD HOLD	Speed hold (SPEED knob)
ALT HOLD	Altitude hold (ALT knob)
HVR HOLD	Hover height hold, HVR ALT HOLD (HVR ALT knob)
AUTO NAC	Automatic nacelles
TF	Terrain following
HVR CPLD	Hover hold
REM HVR	Remote hover: the stick's translation slowed to 2 kt
CRS knob	NAV 1's course
HDG knob	Heading bug
SPEED knob	Speed, KIAS
ALT knob	Altitude, ft
HVR ALT knob	Hover and TF height above the ground, ft
BRT knob	Panel brightness

4.3 Mode annunciation

The mode line along the top of the flight format shows the modes, green when coupled, white as cues: CPLD; the lateral mode (HDG 270, WYPT and the next waypoint, the approach line's name and ARM until it captures); the vertical (ALT 2000, GP ARM, GP 3.0, TF 300 MED); the speed (SPD 120); AUTO NAC; the approach's phase (APPR 4.2, APPR HOVER, APPR LAND, APPR DOWN; APPR HOVER GEAR: the landing armed but the gear up, the hover held until it is down); the hover hold (HVR CPLD DECEL, HOLD, TRC, REM; RAD ALT 50); both engines out, ENG OUT's phase (ENG OUT DIVE, ENG OUT GLIDE 200, ENG OUT FLARE, ENG OUT LAND, ENG OUT DOWN; GEAR after it with the gear up below 1000 ft or in the flare). Over it for 5 s: CPLD OFF, HVR CPLD OFF (let go), HVR CPLD UNAVAIL (refused), AUTO NAC OFF (also alerts, Section 3.4); in it for 5 s, a mode refused: APPR UNAVAIL, VOR/ILS UNAVAIL, TF UNAVAIL.

4.4 The modes

4.4.1 Heading hold (HDG HOLD, `Ctrl-H`)

Turns to the heading bug (magnetic) and holds it, banking at most 15° at 40 kt, 20° at 80 kt and 25° from 150 kt.

4.4.2 Altitude hold (ALT HOLD, `Ctrl-A`)

Climbs or descends to the ALT knob's altitude, at up to 1500 ft/min, and holds it. Engaged in a climb or a descent, it takes the vertical speed there is and changes it at no more than 0.1 *g*.

4.4.3 Speed hold (SPD HOLD, `Ctrl-S`)

Flies the SPEED knob's airspeed, the reference moving to it at 2 kt/s accelerating and 1.5 kt/s slowing, and drives the lever. The reference starts from the aircraft's own airspeed and acceleration and eases into and out of its rate, at no more than the pitch's 0.1 *g* in the exchange of speed for climb (a change of rate of 0.3 kt/s each second at 120 kt, 0.15 at 250 kt), so engaging it slow, or turning the knob, does not pitch the nose. With SPD HOLD on and no vertical mode it holds the vertical speed it had when engaged.

4.4.4 Automatic nacelles (AUTO NAC, `U`)

Sets the nacelles for the speed reference from the aircraft's own level-flight trims: at each speed the most converted angle whose trim pitches the nose up by at most 4° (7° at 180 kt), within the corridor (Figure 5.1). It needs SPD HOLD (AUTO NAC (SPD) without it). Moving the nacelles turns it off (AUTO NAC OFF). With SPD HOLD and AUTO NAC, the SPEED knob alone converts the aircraft either way.

4.4.5 Route (WYPT, `Ctrl-W`)

Flies the route manager's active route: straight to its first waypoint, then between the legs on its own arcs, started where an arc at 80 % of the bank limit joins them (turns of up to 150°). It flies the plan the NAV format draws: APPR arms with the route, so that from the DECEL point it slows down along the route, descends on the approach path and comes to a hover over the route's last point; APPR again lands there (Section 4.4.7). WYPT (NO ROUTE) shows without an active route.

4.4.6 VOR/ILS (`Ctrl-N`)

Flies NAV 1's localizer and glide slope, or its VOR on the CRS knob's course, as a line through the station. The line arms (ILS 28R ARM) until the aircraft is within 20 s of flight of it, the heading or the route flying meanwhile; with neither on it captures at once. It intercepts at up to 45°. The glide path arms (GP ARM) and is joined from below, or from above once past where its descent begins.

4.4.7 Approach (APPR, `Ctrl-G`)

An approach to a hover 50 ft over a point, and a landing there:

- an ILS's threshold, NAV 1 tuned to it;
- the route's destination runway;
- the route's last waypoint, a landing zone, approached into the wind at 6°.

Along the line it slows down at 1.5 kt/s, as late as it can, and converts with AUTO NAC. Below 35 kt over the ground within 2000 ft of the point it hands over to the hover hold, which holds the hover (APPR HOVER). The key again lands (APPR LAND): the height comes down at 2 ft/s and on the ground the modes go off and the lever comes down (APPR DOWN); pressed before the hover it lands without stopping (LAND ARM). A third press turns it off. Without a line to fly it shows APPR UNAVAIL.

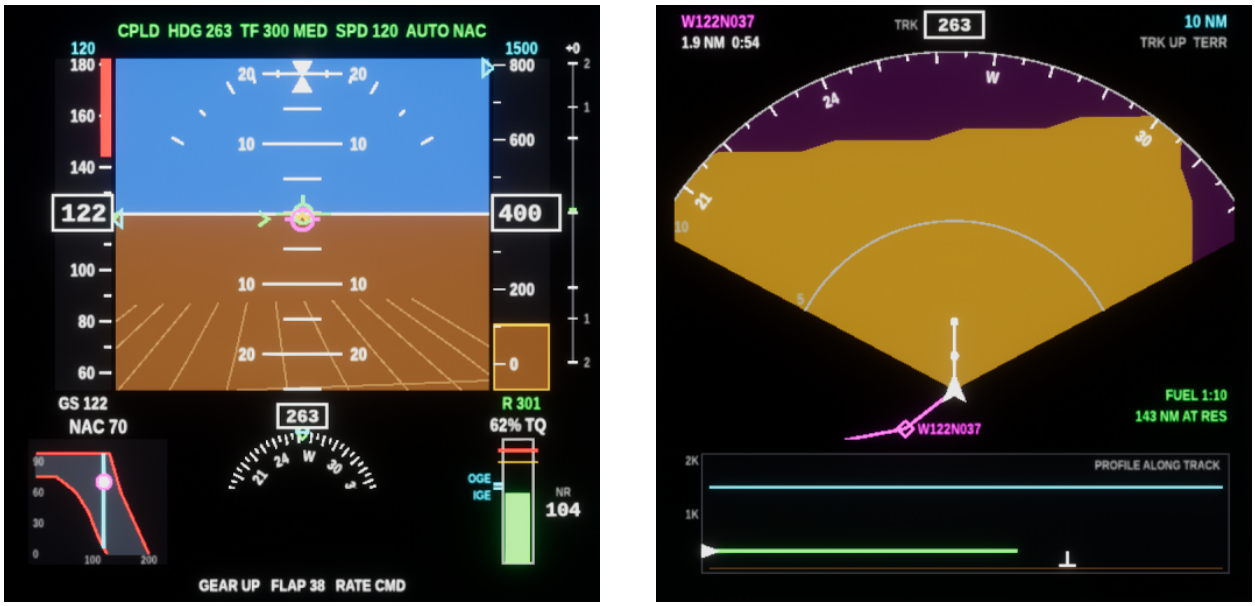
4.4.8 Go-around (GA, `Ctrl-U`)

Ends the approach and the hover hold and flies away: the present heading, the ALT knob's altitude (or 1000 ft up if it is not at least 200 ft above), at the SPEED knob's speed or 80 kt, with AUTO NAC. GA shows until within 100 ft of the altitude.

4.4.9 Terrain following (TF, `Ctrl-T`)

Holds a set clearance over the terrain: the HVR ALT knob's height (200 ft if it is below 100), or the CDU's `T300` (100–1000 ft). It engages from 40 kt over the ground in the air, and turns APPR off; an approach armed later takes the vertical over from it once its glide path is captured.

It plans a path 45 s ahead along the track, the lowest that clears the terrain by the clearance, on the track and 150 ft either side, descending into valleys and pulling up as late as its ride allows, within the climb the engines can make and 2000 ft/min of descent. The ride sets its vertical acceleration (`R1`, `R2`, `R3` on the CDU: soft, medium, hard, 0.1, 0.2 and 0.3 *g*; medium at the start). It flies the vertical: with SPD HOLD the speed is free within 20 kt of the SPEED knob's, so that climbs and descents trade it against height, and held at the band's edges. The mode line shows TF 300 MED.



(a) the flight format

(b) the navigation format, the terrain's profile below

Figure 4.1: Terrain following at 300 ft over the peninsula's ridge, at 120 kt.

Below half the clearance it climbs at full power at 0.5 *g* (FLY UP); with no terrain data ahead it climbs too (TF FAIL). Pressed again TF ends and the ALT knob takes the present altitude.

WARNING

TF sees terrain 45 s ahead along the track: enough to climb 990 ft at the aircraft's weakest climb, 1 440 ft/min at 150 kt. Higher or nearer terrain, terrain coming into view round a turn, and obstacles the scenery's terrain does not have can be cleared only as the fly-up allows.

4.4.10 Hover hold (HVR CPLD, y)

Engaged at up to 40 kt over the ground with the nacelles past 70°, it shows where the aircraft will stop (the magenta box), stops there at 0.08 *g* (DECEL) and holds the spot, the heading and the height (HOLD): the radar height below 500 ft, the barometric above. It moves the lever and the nacelles itself, translating on the nacelles with the fuselage level.

- The stick out of its centre flies it over the ground (TRC): 10 kt at half travel, 2 kt with REM HVR. On release it stops and holds again.
- The lever flies through the height hold; left alone for 1.5 s, the hold takes the new height.
- Past half the stick's travel, above 45 kt, below 65° of nacelle or on landing it lets go (HVR CPLD OFF); refused, it shows HVR CPLD UNAVAIL.

4.4.11 Hover height hold (HVR HOLD, Y)

Holds the HVR ALT knob's radar height, taking the present one when pressed (RAD ALT 50).

4.4.12 Both engines out (ENG OUT)

Comes on when both engines stop in the air; `Ctrl-E` cuts both off, to try it. It flies the landing on the wheels that the model makes best, found by flying it offline from cruise, from airplane mode, from the conversion and from hovers (the procedure, Section 8.2). With CPLD on it flies it; without, its cues are on the flight format and the mode line is white. The other modes go off, the hover hold with them; HDG HOLD stays, and the mode holds its bug (or, without it, the heading it came on at). Its phases:

- DIVE: the lever down, the nacelles commanded to airplane mode (the corridor lets them follow the speed), the nose down for the glide's speed; the descent held to what a pull-out can stop by the flare's height.
- GLIDE 200: the speed held on the pitch.
- FLARE: from 250 ft over the ground (higher when sinking fast), the descent eased to a float 10 ft up, the nose no higher than 6°, the nacelles commanded up as the corridor lets them as the speed falls.
- LAND: the nose at its limit and sinking, the aircraft settles at 3 ft/s, the lever coming up from 20 ft.
- DOWN: on the ground the lever comes down, then the mode goes off.

The gear is the crew's: GEAR on the mode line asks for it below 1000 ft. The panel has no key for it: the name ENGOUT written to the AFCS's `press` property (from a script or a hardware panel) turns it on and off.

4.5 How the modes combine

- *Lateral*: an approach line once captured, else the route (WYPT), else the heading (HDG HOLD). With none the augmentation holds the bank.
- *Vertical*: the glide path once captured, else TF, else ALT HOLD, else the vertical speed held by SPD HOLD.
- *Speed*: SPD HOLD, or the approach's own speed along its path.
- *The hover hold* takes over all of them while it is on; the approach hands over to it.

Part II

Limitations

Chapter 5

Operating limitations

5.1 Conversion corridor

The nacelle angle shall be kept within the corridor (Figure 5.1, Table 5.1); the conversion protection holds the command within it.

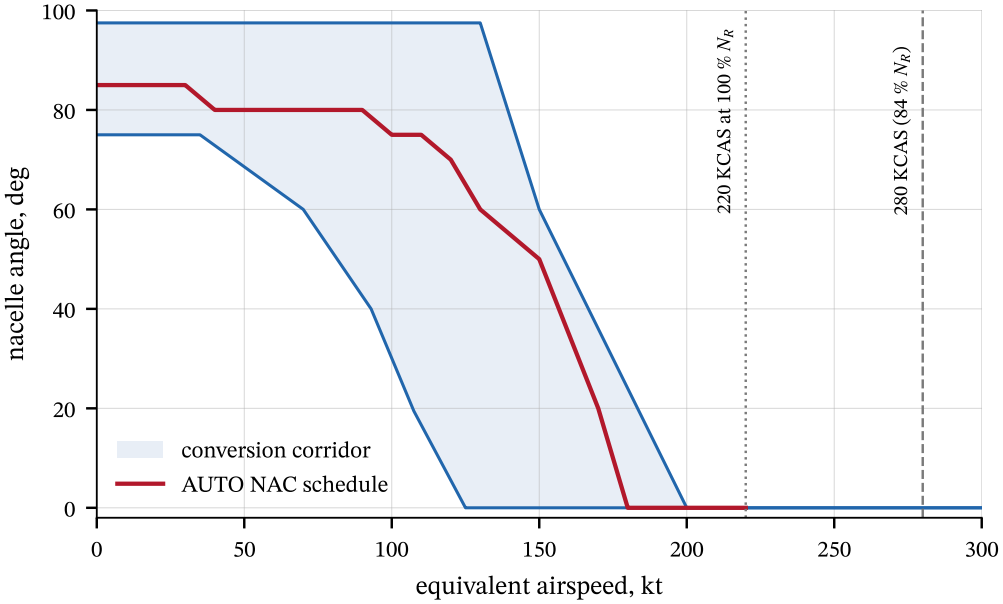


Figure 5.1: The conversion corridor, equivalent airspeed against nacelle angle, with AUTO NAC's schedule and airplane mode's airspeed limits.

Table 5.1: The conversion corridor.

KEAS	least nacelle, deg	most nacelle, deg
0	75	97.5
35	75	97.5
70	60	97.5
93	40	97.5
107	20	97.5
125	0	97.5
130	0	97.5
150	0	60.0
175	0	30.0
200	0	0.0

5.2 Airspeed

- Airplane mode, 84 % N_R: 280 KCAS or Mach 0.48, whichever is less.
- Airplane mode at 100 % N_R: 220 KCAS.
- Gear down: 250 KCAS. Assumed: no figure for the V-22's is public.

5.3 Rotor speed and power

- Rotor speed: 104 % in helicopter mode and through the conversion, 84 % in airplane mode; NR beyond 3 %.
- Torque: each engine's limit, its rating or its share of the drive's limit (9 100 hp at 100 % N_R), whichever is less, marked on its gauge; the engines' controls hold each there.

5.4 High sink rate

WARNING

At low airspeed the rotors can enter their own wake, the vortex ring state, and the aircraft sink fast and roll. The rate of descent shall stay below 800 ft/min up to 30 KCAS, rising to 1 600 ft/min at 60 KCAS (Figure 9.1c). The flight format's vertical speed bar turns amber and red as the descent nears SINK RATE's boundary, which is this limit in vertical descent at the reference loading at sea level and moves with the weight and the air's density.

5.5 Weight

- Maximum vertical take-off: 47 500 lb (the reference loading). The hover ceiling (Part VI) limits it further at altitude.
- Cabin load: 20 000 lb.

5.6 AFCS

- Hover hold: engage below 40 kt over the ground with the nacelles past 70°.
- TF: from 40 kt over the ground; clearance 100–1000 ft.
- Wing stow: rotors stopped.

Part III

Normal procedures

Chapter 6

Normal procedures

6.1 Starting

On the ground the aircraft starts cold and dark. In the air it starts running and trimmed in level flight at the start's airspeed, the nacelles at AUTO NAC's angle for it. For the first 5.5 s it holds still while it trims: the rotors come up to the aircraft's level-flight trim for the airspeed, its weight, the density altitude and the gear (down unless raised, FlightGear's default), the nose turns to the trim's pitch, and the collective, the lever and the stick settle. Then it flies on level, hands off. Slow, heavy and high, past the hover ceiling, the lever is at its stop and the aircraft sinks.

1. Thrust control lever: down.
2. AutoStart: } (or the V22 Osprey menu). The engines start one after the other; the rotor brake releases.
3. Rotor speed: governed (104 %), about 75 s after the first engine lights.

6.2 Vertical take-off and hover

1. Nacelles: 90°. Gear down.
2. Lever: up smoothly until the aircraft lifts (about two thirds at the start's weight).
3. Hover: hold the attitude with the stick; the heading holds with the pedals centred.
4. To hold the hover: HVR CPLD (y); HVR HOLD (Y) holds the radar height.

Note

The flight format's power bar marks the power to hover in and out of ground effect, the model's own for the weight and the air: with the lever's power past the out-of-ground-effect mark the aircraft cannot hover clear of the ground. HOVER PWR warns of it, slowing with the nacelles up, at the height the aircraft is at.

6.3 Conversion to airplane mode

1. Accelerate with forward stick, nacelles at 75–80°.
2. Nacelles: forward (m); the protection holds them to the corridor as the speed builds, so the command may be set fully forward.
3. Gear up above 50 ft.
4. At 0° the rotor speed comes down to 84 %.

Coupled: SPD HOLD and AUTO NAC, then the SPEED knob to the speed wanted; the AFCS converts at constant altitude with ALT HOLD.

6.4 Airplane mode

Cruise 240–270 KCAS. At the slow end of airplane mode the nose is high (6° at 200 KCAS against 3° at 250, Table 7.1): the wing lifts the aircraft alone there.

6.5 Conversion to helicopter mode and landing

1. Slow down below 200 KCAS; nacelles up (`M`) as the corridor allows; gear down below 250 KCAS.
2. Approach with the nacelles at 75–85°, below the high sink rate limit.
3. Come to a hover over the point, then lower the lever to land.
4. After landing: lever down.

6.6 Coupled approach to a hover and landing

1. An ILS: NAV 1 tuned to it; or a route whose last point is the landing zone (WYPT).
2. CPLD; HDG HOLD or WYPT, ALT HOLD, SPD HOLD, AUTO NAC.
3. APPR (`Ctrl-G`): the line arms and is drawn on the NAV format; it captures within 20 s of flight of it.
4. The AFCS descends on the path, slows down and hands over to the hover hold 50 ft over the point.
5. The gear down (`G`).
6. APPR again: it lands, and the lever comes down on the ground. With the gear up it holds the hover (APPR HOVER GEAR) and lands once the gear is down.

With a route flown by WYPT, APPR arms itself.

6.7 Terrain following

1. CPLD; HDG HOLD or WYPT; SPD HOLD, AUTO NAC.
2. Clearance: the HVR ALT knob, or `T300` on the CDU. Ride: `R1`–`R3`.
3. TF (`Ctrl-T`). Check TF 300 MED on the mode line.
4. To end it: TF again; the ALT knob takes the present altitude, and ALT HOLD holds it.

6.8 Shutdown

1. Lever down; AutoStop: `}`.
2. To stow: wing (`N`) once the rotors have stopped.

Part IV

Flight characteristics

Chapter 7

Flight characteristics

7.1 Hover and low speed

With attitude command the aircraft holds the attitude the stick sets, so it stays where it is put; the drift is flown with small attitude changes. In ground effect it needs less power: the power bar shows both. Descending slowly at low speed, mind the high sink rate boundary.

7.2 The conversion

Hands off, the trim schedule holds the attitude through the conversion; the nacelles move at up to 8°/s. Converting, the aircraft accelerates on the nacelles' tilt rather than on its pitch; the pitch attitude changes a few degrees at each angle (Table 7.1).

7.3 Airplane mode

The wing carries the aircraft above about 180 KCAS; at its slow end the nose is 6° up, at cruise 2–3°.

Table 7.1: Level flight along AUTO NAC's schedule at 47 500 lb, 1000 ft: the power required, the pitch attitude, and whether one engine (5 950 hp there) holds it.

KCAS	nacelle, deg	power, hp	pitch, deg	one engine
40	80	6 921	3.4	not enough
60	80	6 179	1.4	not enough
80	80	5 478	1.3	enough
100	75	5 502	3.6	enough
120	70	6 435	2.4	not enough
135	58	6 928	3.9	not enough
150	50	7 543	3.7	not enough
160	35	6 501	4.8	not enough
170	20	5 363	5.6	enough
180	0	3 668	6.9	enough
200	0	3 598	6.2	enough
220	0	4 214	4.5	enough
250	0	5 446	2.9	enough
270	0	6 471	2.1	not enough

7.4 Under the AFCS

Coupled, the AFCS flies the augmentation's attitudes and their rates, the stick for those rates fed forward from the aircraft's own response, so it turns and pulls up without lag or overshoot. In TF the climbs and descents

trade the speed within its band; over hills higher than the ride lets it follow at high speed it stays above the valleys.

Part V

Emergency procedures

Chapter 8

Emergency procedures

8.1 Engine failure

ENG 1 OUT or ENG 2 OUT. The interconnect shaft drives both rotors from the remaining engine, which gives up to its own rating, 6 150 shp at sea level, against the 9 100 hp the drive passes with both. At the reference loading one engine holds level flight from 80 KCAS in the conversion and in airplane mode up to 250 KCAS, not in the hover (6 921 hp at 40 kt) nor at 120–160 KCAS (Table 7.1).

1. In the hover or at low speed: land, or accelerate to 80–100 KCAS with the nacelles at 75–80° if clear of the ground.
2. In the conversion: continue to airplane mode, or back to 80–100 KCAS.
3. In airplane mode: 180–250 KCAS; land as soon as practicable, converting as the speed allows.

WARNING

Below 80 KCAS and at 120–160 KCAS, at the reference loading, one engine cannot hold level flight.

8.2 Both engines out

DUAL ENG OUT; ENG OUT comes on and flies this, coupled, or shows it on the flight format (Section 4.4.12). It is the landing the model makes best on its wheels (the gear's limit 12 ft/s), with the proprotors and the tail clear of the ground, found by flying it offline (`fdm/deadland.py`):

1. The lever down, to flat pitch: the governor holds the rotors windmilling.
2. The nacelles forward to airplane mode, the nose down for 200 KCAS. The corridor holds the nacelles until the speed comes.
3. Glide at 200 KCAS: about 8.4 to 1 in still air at the reference loading, 1.4 NM for every 1000 ft; lighter, less (7.4 to 1 at 38 700 lb, 1.2 NM); with the gear down 7.1 and 6.2 to 1.
4. The gear down below 1000 ft.
5. At 250 ft flare into a float 10 ft over the ground, the nose no higher than 6°: the fuselage meets the ground at 7.3° on the wheels. The nacelles up as far as the corridor lets them as the speed falls: they must reach about 36° for the proprotors to clear the ground.
6. With the nose at 6° and sinking, let it settle; the lever up from 20 ft.

In the model, at the reference loading over flat ground at sea level, it lands intact from cruise, from a glide at 180–200 KCAS from 1500 ft, from the conversion from 4000 ft and from a hover from about 6000 ft (5500 at best): on the wheels at 7.4–11 ft/s and about 161 kt over the ground, the nose at 6°, the nacelles at 43–47° with the proprotors 3.5–4.4 ft clear, 0.5–0.75 NM past the flare's height. With less height it does not land intact. From a hover below 4000 ft, from the conversion at 2000 ft or from 160 KCAS at 800 ft it strikes at 55–90 ft/s, past what is survivable (42 ft/s). From a hover at 4000–5000 ft it strikes at 36–70 ft/s, survivable or

not depending on where the dive's pull-out leaves the glide at the flare. Flown in FlightGear at its start loading (38 700 lb), both engines cut 4.0 NM short of KSFO's runway 28R at 3000 ft and 200 KCAS, the heading bug steered onto the centreline: on the runway 1160 ft past the threshold at 5 ft/s and 144 kt, the nacelles at 69°, stopped 2450 ft past it.

WARNING

The landing is fast, at about 160 kt over the ground: it needs a runway, or flat open ground. Slower it cannot be: held off at 6° of pitch the wing carries the aircraft only to about 150 kt, and nose-higher the tail strikes.

8.3 High sink rate

SINK RATE (its cue NAC FWD POWER): the lever up and the nacelles forward, about 20°. In the model's entries at the reference loading, from the warning this arrests the descent within about 65 ft; from the vortex ring state's onset within about 175 ft and 5 s, where the lever alone takes about 230 ft and the nose down alone does not recover.

8.4 Conversion protection

CONV PROT: the nacelles held off the command by the corridor. Fly the speed into the corridor for the angle commanded, or bring the command to the nacelles.

8.5 AFCS

- CPLD OFF: uncoupled by the stick. Fly the aircraft; CPLD to couple again.
- HVR CPLD OFF: the hover hold let go. Fly the hover; HVR CPLD again once within its limits.
- FLY UP: TF below half its clearance, climbing at full power. Let it climb; if it persists, climb to a safe altitude and end TF.
- TF FAIL: no terrain data ahead; TF climbs. End TF and fly at a safe altitude.

Part VI

Performance data

Chapter 9

Performance

The flight model's own; its report compares it with the published data.

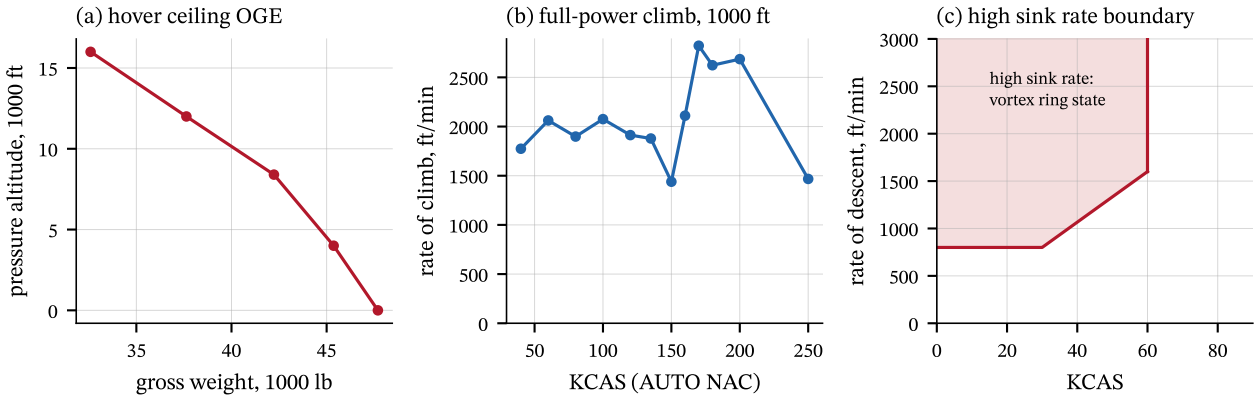


Figure 9.1: (a) The hover ceiling out of ground effect, standard day, 100 % N_R; (b) the climb at full power along AUTO NAC's schedule at 47 500 lb, 1000 ft; (c) the high sink rate boundary.

Table 9.1: Hover ceiling OGE, standard day.

Pressure altitude, ft	Gross weight, lb
0	47 690
4 000	45 370
8 400	42 230
12 000	37 630
16 000	32 600

Table 9.2: Maximum level speed, airplane mode.

Gross weight, lb	Altitude, ft	KTAS
42 000	0	293
47 500	0	291
42 000	17 000	319

Appendix A

Keys

Table A.1: Keys.

Key	Function
D	Cargo doors
d	Crew door
e	Steps door
l	Landing light
N	Fold wing back
n	Unfold wing
m	Tilt Rotors fwd
M	Tilt Rotors aft
j	Jump signal
f	Trigger flare release
q	Refuelling Probe
w	PA On/Off
Q	Reset view
}	Autostart/AutoStop
y	Hover hold (HVR CPLD) on / off
Y	Hover height hold (HVR ALT HOLD) on / off
Ctrl-H	Autopilot heading hold (HDG HOLD)
Ctrl-A	Autopilot altitude hold (ALT HOLD)
Ctrl-S	Autopilot speed hold (SPD HOLD)
Ctrl-W	Autopilot route (WYPT)
Ctrl-N	Autopilot NAV 1's ILS or VOR (VOR/ILS)
Ctrl-G	Autopilot approach to a hover (APPR); again: and land; again: off
Ctrl-T	Autopilot terrain following (TF), at the HVR ALT knob's clearance
Ctrl-E	Test: both engines out (cut off), the battery left on; ENG OUT comes on in the air
Ctrl-U	Autopilot go-around (GA)
u	Couple the autopilot (CPLD)
U	Automatic nacelles (AUTO NAC)
Page Up / Page Down	Thrust control lever up / down (or the throttle axis)
arrow keys, keypad	Stick (elevator and aileron axes)
keypad 0 / Enter	Pedals (rudder axis)
g / G	Landing gear up / down
B	Parking brake